



सो, यू ऑल माइट नो मैथमेटि...
TXT 773.41KB

make notes for me out of this transcript , explain everything properly
deeply

Complete Notes: Statistics & Probability for Data Science

PART 1: DATA VISUALIZATION

1.1 What is Data?

- **Data** = A set of observations/values collected together
- Used in Machine Learning, Deep Learning, Data Mining, etc.
- Stored in **Data Tables**

1.2 Data Tables

- **Tabular representation** of data
- Contains **cells** arranged in **Rows** (horizontal) and **Columns** (vertical)

Example Table:

Name	Age	Salary	Gender
Aakarsh	22	1.2L	Male
Sarthak	25	1.8L	Male
Harsh	28	2.0L	Male
Vedant	24	1.5L	Female

1.3 Individual vs Variable

Term	Definition	Example
Individual	The entity on which dependencies exist	Name (Person)
Variable	Depends on the individual	Age, Salary, Gender

- When you ask "What is the age of Sarthak?" — Sarthak is the **Individual**, Age is the **Variable**
- Every variable depends on the individual

1.4 One-Way Table

- **Definition:** A table with **only ONE individual** and all other columns are **variables** dependent on that individual
- Counter-question test: "Of which person?" → If only one answer possible → One-way table

Why called "One-Way"?

- Because there is only ONE individual (e.g., Name)
- Everything else depends on that one individual

1.5 Two-Way Table

- Has **TWO individuals** (dependencies)
- Counter-questions: "Which month?" AND "Which year?"

Example:

Month	2020	2021	2022	2023
Jan	50k	60k	70k	80k
Mar	80k	85k	90k	95k
May	70k	75k	80k	85k

- **One-way table is INSIDE a two-way table** (e.g., only 2020 data = one-way)

1.6 Types of Variables

Type	Definition	Examples
Quantitative	Continuous numeric data	Age, Salary, Height
Qualitative	Categorical data	Gender, Genre, Colors

1.7 Bar Graph**Steps to create:**

1. Create a **Frequency Table** (count occurrences)
2. Draw horizontal lines (0, 1, 2, 3, 4, 5, 6...)
3. Create bars according to frequency

Example:

Fruit	Frequency
Apple	6
Banana	4
Kiwi	2
Mango	2

- **Most liked fruit** = Apple (tallest bar)
- **Least liked** = Kiwi & Mango (shortest bars)

1.8 Pie Chart

- Uses **frequency counts** but converts to **percentages**
- Formula: $(\text{Frequency} / \text{Total}) \times 100$

Example:

Fruit	Frequency	Percentage
Apple	6	42.9%
Banana	4	28.6%
Kiwi	2	14.3%
Mango	2	14.3%

1.9 Line Graph

- **Purpose:** Show **trends** over time
- Created by plotting dots on top of bars and connecting them
- Then remove the bars

When to use:

- Bar Graph → Categorical comparison
- Line Graph → **Continuous data trends** (e.g., monthly expenses)

1.10 Ogive (Cumulative Frequency Graph)

- Created from a **Cumulative Frequency Table**
- Values keep adding up (cumulative sum)
- Always **increasing** curve

Example:

Month	Expense	Cumulative
Jan	10	10
Feb	12	22
Mar	18	40
Apr	12	52

1.11 Dot Plot

- Stacking dots on top of each other
- Used as a **base for understanding Histograms**

1.12 Histogram

- Looks like a bar graph BUT groups data into **bins/buckets**
- **Steps:**
 1. Find **Range** = Largest - Smallest
 2. Decide number of **bins**
 3. Calculate **bin width** = Range / Number of bins
 4. Create bins and place data points

Example:

- Data: Ages 3 to 24
- Range = $24 - 3 = 21$
- 6 bins $\rightarrow$ Bin width = $21/6 = 3.5 \approx 4$
- Bins: 1-5, 5-9, 9-13, 13-17, 17-21, 21-25

1.13 Relative Frequency Table

- Converts frequencies to **percentages/proportions**
- **Column-related:** Each column totals to 100%
- **Row-related:** Each row totals to 100%

1.14 Joint Distribution

- Compares **two different distributions**
- All percentages total to 100%

PART 2: DATA ANALYSIS

2.1 Measures of Central Tendency

Mean (Average)

- **Formula:** $\bar{x} = (1/n) \times \sum(x_i)$
- Sum of all values divided by total number of values

Population Mean: μ (mu)

Sample Mean: $\bar{x}$ (x-bar)

Example: Data: 2, 4, 7, 9, 10

- Sum = 32
- Mean = $32/5 = 6.4$

Median

- **Middle value** when data is sorted
- **Odd number of data points:** Direct middle value
- **Even number of data points:** Average of two middle values

Example: 4, 7, 9, 10 $\rightarrow$ Median = $(7+9)/2 = 8$

Mode

- **Most frequently occurring value**
- **No mode:** All values distinct
- **Bimodal:** Two modes
- **Multimodal:** More than two modes

Example: 2, 6, 8, 9, 0, 10, 10, 10 $\rightarrow$ Mode = 10

2.2 Measures of Spread

Range

- **Formula:** Range = Largest - Smallest
- Tells the spread of data

Example: 12, 16, 32, 19, 126 → Range = $126 - 12 = 114$

Quartiles (Q1, Q2, Q3)

- **Q1** = 25th percentile
- **Q2** = 50th percentile (Median)
- **Q3** = 75th percentile
- **Q4** = 100th percentile

How to find:

1. Divide data into two equal halves → Q2
2. Divide each half into two equal halves → Q1 and Q3

IQR (Interquartile Range)

- **Formula:** $IQR = Q3 - Q1$
- Measures spread of middle 50% of data

Interpretation:

- Low IQR → Data close to median
- High IQR → Data spread out

2.3 Outliers

Definition: Data points that are unusually far from other observations

Effect of Outliers:

- **Mean** is heavily affected
- **Median** is NOT affected (depends on position)

Example:

- Data: 1, 2, 4, 7, 9, 10, 1001
- Mean = 147.7 (affected by outlier)
- Median = 7 (not affected)

Removing Outliers:

- **Upper Fence** = $Q3 + 1.5 \times IQR$
- **Lower Fence** = $Q1 - 1.5 \times IQR$
- Values outside these fences = Outliers

2.4 Five Number Summary

1. **Minimum**
2. **Q1** (First Quartile)
3. **Median** (Q2)
4. **Q3** (Third Quartile)
5. **Maximum**

Used to create **Box Plots**

2.5 Box Plot

- Visual representation of Five Number Summary

- Outliers shown as **diamonds** outside the whiskers

📌 PART 3: PROBABILITY

3.1 Basics

- **Probability** = Likelihood of an event occurring
- Always between **0 and 1**
- Can be represented as percentage: $P \times 100$

Formula:

$$P(\text{Event}) = \frac{\text{True Outcomes}}{\text{Total Outcomes}}$$

Examples:

- Coin flip: $P(\text{Head}) = 1/2 = 50\%$
- Dice roll: $P(\text{Six}) = 1/6$
- Deck of cards: $P(\text{Heart}) = 13/52 = 1/4 = 25\%$

3.2 Types of Events

Independent Events

- Occurrence of one **does NOT affect** the other
- **Formula:** $P(A \cap B) = P(A) \times P(B)$
- Example: Coin tosses, Dice rolls

Dependent Events

- Occurrence of one **affects** the other
- **Formula:** $P(A \cap B) = P(A) \times P(B|A)$
- Example: Drawing cards without replacement

Mutually Exclusive Events

- **Cannot happen at the same time**
- **Formula:** $P(A \cap B) = 0$
- Example: Getting Head AND Tail in single coin toss

3.3 Probability Rules

Addition Rule (OR)

- **General:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- **Mutually Exclusive:** $P(A \cup B) = P(A) + P(B)$

Multiplication Rule (AND)

- **Independent:** $P(A \cap B) = P(A) \times P(B)$
- **Dependent:** $P(A \cap B) = P(A) \times P(B|A)$

3.4 Conditional Probability

- **Formula:** $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- Probability of A given B has already occurred

Example:

- Contingency Table:

	Ship	Plane	Total
Like Ship	2	4	6
Don't Like Ship	5	3	8
Total	7	7	14

- $P(\text{Likes Ship} \mid \text{Likes Plane}) = 2/7$

3.5 Bayes' Theorem

- Formula:** $P(A|B) = [P(B|A) \times P(A)] / P(B)$
- Used in **Naive Bayes Algorithm** in ML

Example:

- Fair die: $P(6) = 1/6$
- Biased die: $P(6) = 1/2$
- $P(\text{Biased} \mid \text{Rolled } 6) = ?$

PART 4: DATA DISTRIBUTION

4.1 Population vs Sample

Term	Definition	Symbol
Population	Entire group	N
Sample	Subset of population	n

- Population Mean:** μ
- Sample Mean:** $\bar{x}$
- Population Variance:** σ^2
- Sample Variance:** s^2
- Population Standard Deviation:** σ
- Sample Standard Deviation:** s

4.2 Variance

- Measures **spread** of data around mean
- Population:** $\sigma^2 = (1/N) \times \sum (x_i - \mu)^2$
- Sample:** $s^2 = (1/(n-1)) \times \sum (x_i - \bar{x})^2$

4.3 Standard Deviation

- Square root of variance
- Population:** $\sigma = \sqrt{\sigma^2}$
- Sample:** $s = \sqrt{s^2}$

4.4 Density Curve

- Created from Histogram → Frequency Polygon → Density Curve
- **Area under curve = 100% (or 1)**
- **Kernel Density Estimation (KDE) = Density Curve**

4.5 Normal Distribution

- Bell-shaped curve
- Symmetrical
- **Mean = Median = Mode** (all at center)
- Also called **Gaussian Distribution**
- **50% data on left, 50% on right**

4.6 Skewed Distribution

Right Skewed (Positive Skew)

- Tail on right side
- Mean > Median > Mode
- Mean pulled towards tail

Left Skewed (Negative Skew)

- Tail on left side
- Mode > Median > Mean
- Mean pulled towards tail

When to use what:

- **Normal Distribution** → Use Mean
- **Skewed Distribution** → Use Median

4.7 Empirical Rule (68-95-99.7 Rule)

- **68%** of data within **1 standard deviation**
- **95%** of data within **2 standard deviations**
- **99.7%** of data within **3 standard deviations**

Why 99.7% not 100%?

- Normal distribution is **continuous and unbounded** (extends to infinity)
- Tiny probability exists beyond 3 standard deviations

4.8 Z-Score

- **Formula:** $Z = (x - \mu) / \sigma$
- Tells how many **standard deviations** a data point is from the mean
- **Positive Z** → Right side of mean
- **Negative Z** → Left side of mean

Using Z-Table:

- Z-score → Look up in Z-table → Get percentage of data below that point
- Example: $Z = 1.5 \rightarrow 0.9332 = 93.32\%$ data below that point

PART 5: INFERENCE STATISTICS

5.1 Descriptive vs Inferential

Descriptive	Inferential
Summarizes data	Makes predictions
Mean, Median, Mode	Hypothesis Testing
Graphs, Charts	Confidence Intervals
Range, IQR	P-values, Z-test, T-test

5.2 Hypothesis Testing

Steps:

1. **Null Hypothesis (H_0):** No difference/No effect (Innocent until proven guilty)
2. **Alternate Hypothesis (H_1):** There is a difference/effect
3. **Perform Experiment** (Z-test, T-test, etc.)
4. **Accept or Reject H_0**

Significance Value (α):

- Set by domain experts
- Common values: 0.05, 0.01, 0.10
- If not given, assume **0.05**

Confidence Interval:

- Confidence Interval = $1 - \alpha$
- For $\alpha = 0.05 \rightarrow \text{CI} = 95\%$

5.3 Type I and Type II Errors

	H_0 True	H_0 False
Accept H_0	✔ Correct	✗ Type II Error
Reject H_0	✗ Type I Error	✔ Correct

Example (Corona):

- **Type I Error:** Doctor says "You have Corona" but actually you don't
- **Type II Error:** Doctor says "You don't have Corona" but actually you do

5.4 One-Tail vs Two-Tail Test

One-Tail Test	Two-Tail Test
Direction specified ($>$, $<$)	No direction ($\neq$)
α on one side	$\alpha/2$ on both sides
Example: "More than 70"	Example: "Different than 70"

5.5 Z-Test

When to use:

- Sample size > 30
- **Population** standard deviation is **known**

Formula:

$$Z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$$

Critical Values ($\alpha = 0.05$):

- Two-tail: ± 1.96
- One-tail: ± 1.645

5.6 T-Test

When to use:

- Sample size < 30 (usually)
- **Population** standard deviation is **unknown**
- **Sample** standard deviation is known

Formula:

$$t = (\bar{x} - \mu) / (s / \sqrt{n})$$

Degrees of Freedom: $df = n - 1$

5.7 Two-Sample Tests

- Compare **two different samples**
- **Z-test:** When population standard deviations are known
- **T-test:** When sample standard deviations are known

Degrees of Freedom: $\min(n1 - 1, n2 - 1)$

5.8 Chi-Square Test

When to use:

- **Categorical variables**
- Check if there is a **significant association** between two categorical variables

Formula:

$$\chi^2 = \sum [(O - E)^2 / E]$$

Where:

- O = Observed frequency
- E = Expected frequency = (Row Total $\times$ Column Total) / Grand Total

Degrees of Freedom: $(\text{Rows} - 1) \times (\text{Columns} - 1)$

5.9 ANOVA (Analysis of Variance)

When to use:

- Compare means of **more than two independent groups**

F-Test = ANOVA Test

Null Hypothesis: No significant difference between group means

Alternate Hypothesis: At least one group mean is different

5.10 P-Value Approach

- **P-value** = Probability value
- Compare with α (**significance level**)
- If **P-value** < $\alpha \rightarrow$ Reject H_0
- If **P-value** > $\alpha \rightarrow$ Accept H_0

In code:

```
python

if p_value < alpha:
    print("Reject Null Hypothesis")
else:
    print("Accept Null Hypothesis")
```

PART 6: COVARIANCE & CORRELATION

6.1 Covariance

- Tells how two variables change together
- **Positive Covariance:** Both increase together
- **Negative Covariance:** One increases, other decreases

Formula:

$$\text{Cov}(X, Y) = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] / (n - 1)$$

Limitation: Only gives direction, not strength

6.2 Correlation

- Tells **strength** of relationship
- Always between **-1 and +1**

Formula:

$$\text{Corr}(X, Y) = \text{Cov}(X, Y) / (\sigma_x \times \sigma_y)$$

Interpretation:

Value	Meaning
+1	Perfect positive correlation
0	No correlation
-1	Perfect negative correlation
Close to +1	Strong positive
Close to -1	Strong negative
Close to 0	Weak/No correlation

6.3 Use Case in ML

- If a column has **no correlation** with target → Don't use for prediction
- If **high correlation** → Use for prediction

SUMMARY TABLE

Topic	Key Formula/Concept
Mean	$\Sigma x_i / n$
Median	Middle value
Mode	Most frequent
Variance	$\Sigma (x_i - \mu)^2 / N$
Standard Deviation	$\sqrt{\text{Variance}}$
Z-Score	$(x - \mu) / \sigma$
Z-Test	$(\bar{x} - \mu) / (\sigma / \sqrt{n})$
T-Test	$(\bar{x} - \mu) / (s / \sqrt{n})$
Chi-Square	$\Sigma [(O-E)^2 / E]$
Covariance	$\Sigma [(x_i - \bar{x})(y_i - \bar{y})] / (n-1)$
Correlation	$\text{Cov}(X,Y) / (\sigma_x \times \sigma_y)$

KEY TAKEAWAYS

1. **Statistics** is the foundation of Data Science
2. **Visualization** helps understand data before analysis
3. **Mean, Median, Mode** describe central tendency
4. **Range, IQR, Variance, Std Dev** describe spread
5. **Outliers** must be removed before ML modeling
6. **Normal Distribution** is the most important distribution
7. **Empirical Rule:** 68-95-99.7
8. **Z-Score** tells how far a point is from mean
9. **Hypothesis Testing** validates claims about population
10. **Z-test** ($n > 30$, σ known) vs **T-test** ($n < 30$, σ unknown)
11. **Chi-Square** for categorical variables
12. **ANOVA** for comparing 3+ groups
13. **Covariance** gives direction, **Correlation** gives strength
14. **P-value** $< \alpha \rightarrow$ Reject Null Hypothesis
15. **Statistics + Probability** = Foundation for Machine Learning

give me interview questions with answers topic wise of all the things covered in this chat

Statistics & Probability Interview Questions with Answers

TOPIC 1: DATA VISUALIZATION

Q1: What is the difference between a bar graph and a histogram?

Answer:

Bar Graph	Histogram
Used for categorical data	Used for continuous/numerical data
Bars have gaps between them	Bars are adjacent (no gaps)
Each bar represents a category	Each bar represents a bin/range
Example: Fruits vs Frequency	Example: Age groups vs Frequency

Q2: What is a two-way table? How is it different from a one-way table?

Answer:

- **One-way table:** Has only ONE individual (independent variable). Everything depends on that single individual. Example: Name → Age, Salary, Gender.
- **Two-way table:** Has TWO individuals (dependencies). Example: Month AND Year → Revenue.
- A one-way table is actually **inside** a two-way table (e.g., data for only 2020).

Q3: What is the purpose of a line graph vs a bar graph?

Answer:

- **Bar Graph:** Compare categorical data (e.g., fruit frequencies)
- **Line Graph:** Show **trends over time** for continuous data (e.g., monthly expenses)
- Line graphs are created by plotting dots on top of bars and connecting them.

Q4: What is an Ogive?

Answer:

- An **Ogive** is a line graph created from a **Cumulative Frequency Table**.
- It always **increases** (or stays flat) because we keep adding values.
- Used to find how many data points fall below a certain value.

Q5: What is a Dot Plot? Why is it used?

Answer:

- A **Dot Plot** stacks dots on top of each other instead of bars.

- It is a **precursor to Histograms**.
- Helps visualize the distribution of small datasets.

Q6: What is a Relative Frequency Table?

Answer:

- Converts frequencies into **percentages/proportions**.
- **Column-related**: Each column totals to 100%.
- **Row-related**: Each row totals to 100%.
- Used in Machine Learning for **dimensionality reduction** and **encoding categorical variables**.

Q7: What is a Joint Distribution?

Answer:

- Compares **two different distributions** in a single table.
- All percentages total to **100%**.
- Example: Weight Loss vs Total Hours Workout.

TOPIC 2: DATA ANALYSIS

Q8: What is the difference between Mean, Median, and Mode?

Answer:

Measure	Definition	Affected by Outliers?	Best For
Mean	Average ($\Sigma x/n$)	 Yes	Normal di
Median	Middle value	 No	Skewed d
Mode	Most frequent value	 No	Categoric

Q9: When should you use Median over Mean?

Answer:

- Use **Median** when data has **outliers** or is **skewed**.
- Example: Income data (few billionaires skew the mean).
- Use **Mean** when data is **normally distributed** and has no extreme outliers.

Q10: What is IQR? How is it used to detect outliers?

Answer:

- **IQR (Interquartile Range)** = $Q3 - Q1$
- **Upper Fence** = $Q3 + 1.5 \times \text{IQR}$

- **Lower Fence** = $Q1 - 1.5 \times IQR$
- Any data point **outside these fences** is an **outlier**.
- IQR measures the spread of the **middle 50%** of data.

Q11: What is the Five Number Summary?

Answer:

1. Minimum
2. Q1 (First Quartile)
3. Median (Q2)
4. Q3 (Third Quartile)
5. Maximum

Used to create **Box Plots**. Outliers are shown as diamonds outside the whiskers.

Q12: What is a Box Plot? What does it tell you?

Answer:

- A **Box Plot** is a visual representation of the Five Number Summary.
- Shows: Minimum, Q1, Median, Q3, Maximum
- **Outliers** are plotted as individual points (diamonds) outside the whiskers.
- Helps compare distributions across groups.

Q13: How do outliers affect Mean and Median?

Answer:

- **Mean:** Heavily affected (pulled towards outlier)
- **Median:** NOT affected (depends on position, not value)
- Example: Data 1, 2, 4, 7, 9, 10, 1001 → Mean = 147.7, Median = 7
- **Always remove outliers** before ML modeling (especially for regression).

TOPIC 3: PROBABILITY

Q14: What is Probability? What is its range?

Answer:

- **Probability** = Likelihood of an event occurring.
- Range: **0 to 1** (or 0% to 100%).
- Formula: $P(\text{Event}) = \frac{\text{True Outcomes}}{\text{Total Outcomes}}$
- Example: $P(\text{Head}) = 1/2 = 50\%$

Q15: What are Independent Events? Give an example.

Answer:

- Events where the occurrence of one **does NOT affect** the other.
- Formula: $P(A \cap B) = P(A) \times P(B)$
- Example: Coin tosses (first toss doesn't affect second toss)
- Example: Dice rolls

Q16: What are Dependent Events? Give an example.

Answer:

- Events where the occurrence of one **affects** the other.
- Formula: $P(A \cap B) = P(A) \times P(B|A)$
- Example: Drawing cards without replacement
- First draw: $P(\text{King}) = 4/52$
- Second draw: $P(\text{King}) = 3/51$ (if first was King)

Q17: What are Mutually Exclusive Events?

Answer:

- Events that **cannot happen at the same time**.
- Formula: $P(A \cap B) = 0$
- Example: Getting Head AND Tail in a single coin toss
- Example: Rolling 1 AND 6 on a single die roll

Q18: State the Addition Rule of Probability.

Answer:

- **General:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- **Mutually Exclusive:** $P(A \cup B) = P(A) + P(B)$ (since $P(A \cap B) = 0$)
- Used when question says "**OR**" or "**Either**"

Q19: State the Multiplication Rule of Probability.

Answer:

- **Independent:** $P(A \cap B) = P(A) \times P(B)$
- **Dependent:** $P(A \cap B) = P(A) \times P(B|A)$
- Used when question says "**AND**"

Q20: What is Conditional Probability?

Answer:

- Probability of event A **given** event B has already occurred.
- Formula: $P(A|B) = P(A \cap B) / P(B)$

- Example: $P(\text{Likes Ship} \mid \text{Likes Plane}) = 2/7$

Q21: What is Bayes' Theorem? Where is it used?

Answer:

- Formula: $P(A|B) = [P(B|A) \times P(A)] / P(B)$
- Used in **Naive Bayes Algorithm** in Machine Learning.
- Updates probability based on new evidence.
- Example: $P(\text{Biased Die} \mid \text{Rolled 6}) = 3/4$

TOPIC 4: DATA DISTRIBUTION

Q22: What is the difference between Population and Sample?

Answer:

Population	Sample
Entire group	Subset of population
N (total count)	n (sample size)
Mean = μ	Mean = $\bar{x}$
Variance = σ^2	Variance = s^2
Std Dev = σ	Std Dev = s

Q23: What is Variance? How is it calculated?

Answer:

- Measures **spread** of data around the mean.
- **Population:** $\sigma^2 = (1/N) \times \sum (x_i - \mu)^2$
- **Sample:** $s^2 = (1/(n-1)) \times \sum (x_i - \bar{x})^2$
- Note: Sample uses **n-1** (Bessel's correction) to get unbiased estimate.

Q24: What is Standard Deviation? How is it related to Variance?

Answer:

- **Standard Deviation** = Square root of Variance.
- $\sigma = \sqrt{\sigma^2}$ (Population)
- $s = \sqrt{s^2}$ (Sample)
- Represents average distance of data points from the mean.

Q25: What is a Normal Distribution? What are its properties?

Answer:

- **Bell-shaped curve**
- **Symmetrical** (50% left, 50% right)
- **Mean = Median = Mode** (all at center)
- Also called **Gaussian Distribution**
- Area under curve = 100% (or 1)
- Follows **Empirical Rule**: 68-95-99.7

Q26: What is Skewness? What are its types?

Answer:

- **Skewness** = Asymmetry in distribution.
- **Right Skewed (Positive)**: Tail on right, Mean > Median > Mode
- **Left Skewed (Negative)**: Tail on left, Mode > Median > Mean
- In skewed data, **Median** is a better measure of central tendency.

Q27: What is the Empirical Rule?

Answer:

- **68%** of data within **1 standard deviation** from mean.
- **95%** of data within **2 standard deviations**.
- **99.7%** of data within **3 standard deviations**.
- Why not 100%? Because normal distribution extends to **infinity** (tiny probability beyond 3 SD).

Q28: What is a Z-Score? How is it calculated?

Answer:

- Tells how many **standard deviations** a point is from the mean.
- Formula: $z = (x - \mu) / \sigma$
- **Positive Z**: Right side of mean
- **Negative Z**: Left side of mean
- Used with **Z-table** to find percentage of data below a point.

Q29: How do you use a Z-table?

Answer:

1. Calculate Z-score.
 2. Look up Z-score in Z-table.
 3. The value gives the **percentage of data below** that point.
 4. For above: $100\% - \text{value}$.
- Example: $Z = 1.5 \rightarrow 0.9332 = 93.32\%$ data below.

TOPIC 5: INFERENCE STATISTICS

Q30: What is the difference between Descriptive and Inferential Statistics?

Answer:

Descriptive	Inferential
Summarizes data	Makes predictions
Mean, Median, Mode	Hypothesis Testing
Graphs, Charts	Confidence Intervals
Range, IQR	P-values, Z-test, T-test

Q31: What is Hypothesis Testing? What are its steps?

Answer:

1. **Null Hypothesis (H_0):** No difference/No effect (innocent until proven guilty)
2. **Alternate Hypothesis (H_1):** There is a difference/effect
3. **Perform Experiment:** Z-test, T-test, Chi-square, ANOVA
4. **Accept or Reject H_0** based on critical values or p-values

Q32: What is Significance Level (α)?

Answer:

- Probability of rejecting H_0 when it is actually true (Type I Error).
- Set by **domain experts**.
- Common values: **0.05, 0.01, 0.10**
- If not given in question, assume **0.05**.

Q33: What is Confidence Interval?

Answer:

- Range of values that likely contains the population parameter.
- Formula: $CI = 1 - \alpha$
- For $\alpha = 0.05 \rightarrow CI = 95\%$
- If test statistic falls inside CI $\rightarrow$ **Accept H_0**
- If outside CI $\rightarrow$ **Reject H_0**

Q34: What are Type I and Type II Errors?

Answer:

	H_0 True	H_0 False
Accept H_0	✔ Correct	✗ Type II Error
Reject H_0	✗ Type I Error	✔ Correct

- **Type I Error:** False Positive (Rejecting true H_0)
- **Type II Error:** False Negative (Accepting false H_0)
- Example: Corona test — Type I = "You have Corona" (but you don't), Type II = "You don't have Corona" (but you do)

Q35: What is the difference between One-Tail and Two-Tail Test?

Answer:

One-Tail Test	Two-Tail Test
Direction specified ($>$, $<$)	No direction ($\neq$)
α on one side	$\alpha/2$ on both sides
Example: "More than 70"	Example: "Different than 70"
Critical value ($\alpha=0.05$): ± 1.645	Critical value ($\alpha=0.05$): ± 1.96

Q36: When do you use a Z-test vs T-test?

Answer:

Z-Test	T-Test
Sample size > 30	Sample size < 30 (usually)
Population Std Dev known	Population Std Dev unknown
Uses Z-table	Uses T-table
Formula: $Z = (\bar{x} - \mu)/(\sigma/\sqrt{n})$	Formula: $t = (\bar{x} - \mu)/(s/\sqrt{n})$

Q37: What is Degrees of Freedom in T-test?

Answer:

- $df = n - 1$ (for one-sample T-test)
- For two-sample T-test: $df = \min(n_1 - 1, n_2 - 1)$
- Used to find critical values in T-table.
- Represents number of independent pieces of information.

Q38: What is a Two-Sample T-test?

Answer:

- Compares means of **two independent samples**.
- **Z-test**: When population std dev is known.
- **T-test**: When sample std dev is known.
- Formula: $t = (\bar{x}_1 - \bar{x}_2) / \sqrt{[(s_1^2/n_1) + (s_2^2/n_2)]}$

Q39: When do you use Chi-Square Test?

Answer:

- When both variables are **categorical**.
- Checks if there is a **significant association** between two categorical variables.
- Formula: $\chi^2 = \sum [(O - E)^2 / E]$
- Where $E = (\text{Row Total} \times \text{Column Total}) / \text{Grand Total}$
- Degrees of Freedom: $(\text{Rows} - 1) \times (\text{Columns} - 1)$

Q40: When do you use ANOVA?

Answer:

- When comparing means of **more than two groups**.
- ANOVA = Analysis of Variance.
- **F-test** = ANOVA Test.
- Null Hypothesis: No significant difference between group means.
- Alternate Hypothesis: At least one group mean is different.

Q41: What is a P-value? How is it used?

Answer:

- **P-value** = Probability of obtaining results at least as extreme as observed.
- Compare with α (**significance level**).
- If **P-value** $< \alpha \rightarrow$ Reject H_0 (significant result)
- If **P-value** $> \alpha \rightarrow$ Accept H_0 (not significant)
- In code: `if p_value < alpha: reject H0`

Q42: What is the difference between Critical Value and P-value approaches?

Answer:

Critical Value	P-value
Compare test statistic with critical value	Compare p-value with α
Used for manual solving	Used in code implementation
Example: $Z > 1.96 \rightarrow$ Reject H_0	Example: $p < 0.05 \rightarrow$ Reject H_0
Easier for small problems	Easier for large problems/code

TOPIC 6: COVARIANCE & CORRELATION

Q43: What is Covariance?

Answer:

- Measures how two variables change together.
- **Positive Covariance:** Both increase together.
- **Negative Covariance:** One increases, other decreases.
- Formula: $\text{Cov}(X, Y) = \sum [(x_i - \bar{x})(y_i - \bar{y})] / (n - 1)$
- **Limitation:** Only gives direction, not strength.

Q44: What is Correlation? How is it different from Covariance?

Answer:

- **Correlation** = Standardized Covariance.
- Tells **strength** and **direction** of relationship.
- Always between **-1 and +1**.
- Formula: $\text{Corr}(X, Y) = \text{Cov}(X, Y) / (\sigma_x \times \sigma_y)$
- Covariance gives direction only; Correlation gives both direction and strength.

Q45: Interpret Correlation values.

Answer:

Value	Meaning
+1	Perfect positive correlation
0	No correlation
-1	Perfect negative correlation
Close to +1	Strong positive
Close to -1	Strong negative
Close to 0	Weak/No correlation

Q46: How is Correlation used in Machine Learning?

Answer:

- **Feature Selection:** If a feature has no correlation with target → Don't use it.
- If high correlation → Use it for prediction.
- Example: Age and Salary are correlated → Use Age to predict Salary.
- Helps in **dimensionality reduction** and **feature engineering**.

Q47: What is the relationship between Covariance and Correlation?**Answer:**

- $\text{Correlation} = \text{Covariance} / (\sigma_x \times \sigma_y)$
- Correlation is **Covariance normalized** by standard deviations.
- Covariance is **scale-dependent**; Correlation is **scale-independent**.
- Correlation is always between -1 and +1; Covariance has no bounds.

 BONUS: MACHINE LEARNING CONNECTION**Q48: Where is Bayes' Theorem used in Machine Learning?****Answer:**

- **Naive Bayes Algorithm** (for classification)
- Spam filtering
- Sentiment analysis
- Document classification

Q49: Why do we remove outliers before ML modeling?**Answer:**

- Outliers **distort mean and standard deviation**.
- They **disproportionately influence** regression models.
- Lead to **poor generalization** and bad predictions.
- Remove using **IQR method** (Upper/Lower Fence).

Q50: Why is Statistics important for Data Science?**Answer:**

- Foundation for **Machine Learning algorithms**.
- Helps in **data cleaning** and **preprocessing**.
- Used in **hypothesis testing** for model validation.
- Essential for **feature selection** (correlation, covariance).
- Helps in **understanding data distribution** before modeling.

 QUICK REVISION TABLE

Topic	Key Formula/Concept
Mean	$\Sigma x_i / n$
Median	Middle value
Mode	Most frequent

Topic	Key Formula/Concept
Variance	$\sum (x_i - \mu)^2 / N$
Standard Deviation	$\sqrt{\text{Variance}}$
Z-Score	$(x - \mu) / \sigma$
Z-Test	$(\bar{x} - \mu) / (\sigma / \sqrt{n})$
T-Test	$(\bar{x} - \mu) / (s / \sqrt{n})$
Chi-Square	$\sum [(O-E)^2 / E]$
Covariance	$\sum [(x_i - \bar{x})(y_i - \bar{y})] / (n-1)$
Correlation	$\text{Cov}(X,Y) / (\sigma_x \times \sigma_y)$
Bayes' Theorem	$P(A B) = [P(B A) \times P(A)] / P(B)$

FINAL TIPS FOR INTERVIEW

1. **Always ask clarifying questions** — Is σ known? Sample size? Categorical or continuous?
2. **Know when to use what** — Z-test vs T-test vs Chi-square vs ANOVA
3. **Understand the "why"** — Not just formulas, but intuition behind them
4. **Practice with real datasets** — Titanic, Iris, etc.
5. **Connect to ML** — How each concept applies to Machine Learning
6. **Be ready to code** — Python (NumPy, SciPy, Pandas, Seaborn)

Good luck with your interview! 🚀